

2023 Ph H1 Q2

Section: Our Dynamic Universe

Topic: Motion, Equations and Graphs

Question Summary

A sandbag is released from 50 m height, taking 3.0 s to reach the ground (ignore air resistance).

What is its initial velocity?



Final Answer:

B. 2.0 m s^{-1} downwards

Working

$$s = ut + \frac{1}{2}at^2$$

$$-50 = 3u - 44.1$$

$$3u = -5.9 \Rightarrow u = -1.97 \text{ m s}^{-1}$$

Downwards $2.0 \text{ m s}^{-1} \rightarrow \text{B.}$

Quick Tips

- Choose a consistent sign convention (up = +).
- Negative u means downward motion.